

Sextuple pairs with square denominators

Problèmes ouverts — A. Dujella · 2026

Problem 3.12 (Dujella, 2026). Are there infinitely many pairs of rational Diophantine sextuples sharing a common triple such that denominators of all elements (in the lowest terms) in the sextuples are perfect squares?

Remark. In 2024, Rathbun found two pairs of such sextuples. One such pair is

$$\{3/2^2, -12/5^2, -127500/389^2, -2093/50^2, 72512/265^2, -271859133/27970^2\},$$

$$\{3/2^2, -12/5^2, -127500/389^2, 2387/50^2, 435252/685^2, -38178413/6130^2\}.$$